Pricing Off Accounting Costs

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Pricing and Inventory Purchase Costs: Fed Beige Book

- "Several contacts noted they were still working through pre-tariff inventories, thus delaying price adjustments."

  — Federal Reserve Bank of Atlanta, June 2025

- "Several contacts across different retail sectors phased in price increases as old inventories were sold and replaced with goods subject to tariffs. Retailers expect additional price increases as this phase-in continues."

  — Federal Reserve Bank of San Francisco, Sep 2025

- "A coffee roaster noted that while tariffs on coffee have largely been lifted, selling prices will only go down once the stock of inventory acquired at higher costs has been cleared."

  — Federal Reserve Bank of New York, Jan 2026

Pricing and Inventory Purchase Costs

- Standard theory: price off marginal cost = current replacement cost.
- Historical purchase cost is sunk and thus irrelevant!
- Yet, firms widely report setting prices based on past inventory purchase costs.

Pricing and Inventory Purchase Costs

- Standard theory: price off marginal cost = current replacement cost.
- Historical purchase cost is sunk and thus irrelevant!
- Yet, firms widely report setting prices based on past inventory purchase costs.
- Part of broader pattern of behavior:
  - Firms' prices affected by hedging gains/losses: Evidence from airlines. [in paper]
  - Firms report passing through changes in fixed costs.
  - Little response to current/future replacement costs, demand shocks.

Why?

Retailer CEO: All tariff pauses just ended on August 8. . . we all brought in as much previously (lower) tariffed goods before that, and most fall merchandise was here in the spring and early summer. . . so we are still selling goods that didn't have the latest tariff burden. But we are all raising prices slowly and carefully as the tariffed inventory gets sold (remember, accounting wise, we have a cash hit paying tariffs upon receipt, but we have a P&L hit when we sell the tariffed goods). [....]

Cochrane: It's interesting how the accounting treatment of bottom line matters. A pure economist would immediately mark up the current inventory to its opportunity cost[...], not its historical cost, book a windfall profit, and then mark any sales below that value as losses. Business[es] only raise prices when accounting losses force them to do so.

Roadmap

1. Principal–agent model of pricing off accounting costs.
   - Replacement costs unverifiable, but accounting cash flows verifiable and contractible.
   - Result: Firms set weighted avg. of “replacement cost” and “accounting cost” prices.
   - $\downarrow$ accounting cost pricing with high competition, observable costs.
   - Other puzzles: pass-through of fixed costs, muted response to demand shocks.

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   - Other puzzles: pass-through of fixed costs, muted response to demand shocks.
2. Estimate the extent of accounting cost pricing using financial reporting data.
   - Weight on accounting costs $=$ 60–80%.
   - Replacement cost pricing largely firms in competitive industries / with observable costs.

Roadmap

3. Dynamics under accounting cost pricing resemble sticky-price models.
   - But lags come from production lags and inventory stocks, not length of price spells.
   - More delayed response to slow-moving shocks, more amplification in supply chains.
   - Anticipation can further delay pass-through if firms stock up on inventory.
4. How far can we go without sticky prices?
   - Calibrate U.S. input–output model.
   - Delays in CPI adjustment to shocks on par with sticky-price models.

Selected Related Literature

- Cost-based pricing: Gordon (1981), Blinder et al. (1998).
  - Firms inattentive to / imperfectly observe future costs: Gordon (1990), Mankiw and Reis (2002), Woodford (2003), Minton and Wheaton (2022), Afrouzi et al. (2024).
  - Okun (1981), Gordon (1981): "Many firms, however, appear to base prices on actually costs paid in the past, so-called 'FIFO' (first-in-first-out) pricing practices."
  - Fairness: Kahneman et al. (1986), Rotemberg (2005, 2011), Eyster et al. (2021).
- Other pricing puzzles:
  - Full-cost pricing: E.g., Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
  - Response to cost vs. demand shocks: Cagan (1979), Bils and Chang (2000), Fabiani et al. (2006), Gagnon and Lopez-Salido (2020), Kohler et al. (2026).
- Verifiable info and monitoring: E.g., Meckling and Jensen (1976), Holmström (1979), Graham et al. (2005), Rogerson (2008, 2011).

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Pricing and inventory purchase costs: Earnings calls

- Retailers: “Our pricing algorithms seek to maintain a balance between factoring in our higher wholesale costs and keeping us market priced. In practice, this means that for a short period of time, inventory-carrying retailers may have a slight cost advantage. This should dissipate over the next couple of quarters as the market rebalances, pre-tariff inventory is depleted, and all retailers incur higher wholesale costs reflective of tariffs.”

  — Wayfair, Oct 31 2019

- Distributors: “We anticipate implementing additional pricing actions as inventory affected by tariffs moves through our cost of sales.”

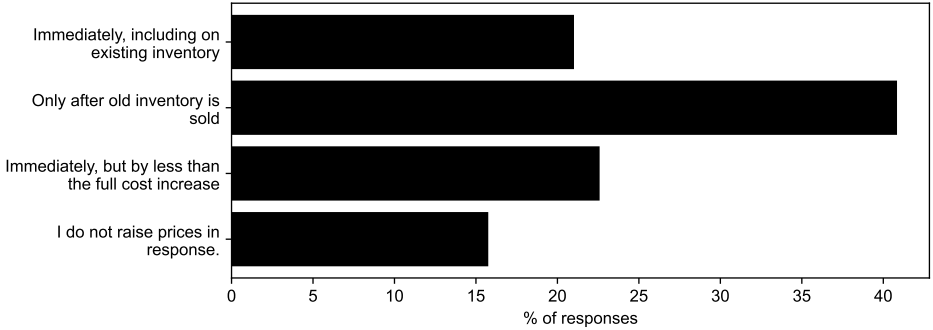
  — Global Industrial, July 29 2025

- Manufacturers: “We have the opportunity for those duties that we pay, those tariffs that we pay, to recover that when we deliver the aircraft. Now, you know, it creates a little bit of a cash flow timing issue that we’ll have to work through.”

  — Boeing, Apr 23 2025

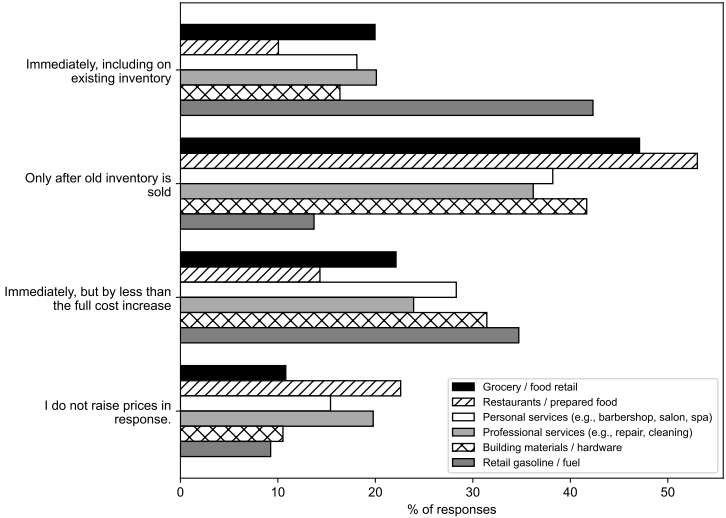
Survey evidence

- "Suppose your input cost rises today, but you still have inventory bought at lower cost. When do you raise your sales price?"



Note: Survey of 400 firm price-setters by Schoenle (2026).

Exceptions



- Exception: gas stations.
- "The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal." (Okun 1981).
- Another (anecdotal) exception: sellers of gold/silver jewelry.

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Stripped-down, two-period model

$t = 0$	$t = 1$	$t = 2$
Manager purchases I units of inventory at price h.	Manager observes c and sets p. Principal chooses payment K given sales $pD(p)$ and COGS $hD(p)$.	Manager liquidates leftover inventory at price c.

- No discounting. Assume I fixed and $D(p) < I$ for all possible p (no stockouts).
- $V^{\text{FB}} = \max_p \{-hI + pD(p) + c(I - D(p))\}$.
- $p^* = \arg\max_p (p - c)D(p)$.

Two-period model: Principal–agent problem

- Principal delegates firm operations to manager.

$$V^{\text{manager}} = \max_{p} \Big\{ \underbrace{-hI + pD(p) + c(I - D(p))}_{\text{True firm value}} + \underbrace{(\alpha - 1)p}_{\text{Pricing wedge}} + \underbrace{K(pD(p), hD(p))}_{\text{Incentive payment}} \Big\}.$$

- Conflicting incentives when $\alpha \neq 1$.
  - Wants to set price lower or higher than optimal.
    E.g., Baumol (1959), Bertrand and Mullainathan (2003), Graham et al. (2005).

- Both manager preference α and replacement cost c are private information.

- Principal can design a payment at $t = 1$ to incentivize manager behavior:

$$V^{\text{principal}} = \max_{K(\cdot)} \mathbb{E}\left[-hI + pD(p) + c(I - D(p))\right].$$

Two-period model: Principal–agent problem

- Why can't manager just tell principal c?
  - Manager always has incentive to misreport c in order to justify desired price.
- Why can't principal condition payment on liquidation proceeds at $t = 2$?
  - Liquidation proceeds reveal c to both parties.
  - We assume that contract must be tied to performance observable at $t = 1 \Rightarrow$ need to monitor and incentivize manager in a timely manner.
- Principal's trade-off:
  - Stronger incentives limit extent to which manager preference α affects price...
  - ...but also lead manager to disregard valuable info about c.

Two-period model: Principal–agent problem

- For analytic results, several parametric assumptions.
- Isoelastic demand: $D(p) = p^{-\eta}$.
- Assume principal's payment has quadratic form around target accounting markup μ,

$$K(pD(p), hD(p)) = -\frac{\kappa}{2}\left(\frac{pD(p)}{hD(p)} - \mu\right)^2.$$

- Define replacement cost shock $\varepsilon = c/h$.
- Log-linearize around $h = c = 1$, $\alpha = 1$. Let $\hat{x}$ be log-deviation of x from value at this pt.

Pricing without incentives

- Suppose $\kappa = 0$. Then, price set by manager is

$$\hat{p} = \hat{c} + \frac{\eta^{\eta}}{(\eta - 1)^{\eta+1}} \hat{\alpha}.$$

- Elasticity of price to manager preference decreasing in demand elasticity η.
- Firm value more convex in price when elasticity of substitution high.
- $\Rightarrow$ Managerial slack harder to sustain in competitive industries.

  Hart (1983), Giroud and Mueller (2010, 2011).

Optimal contract parameters

- Given quadratic contract form,

$$K(pD(p), hD(p)) = -\frac{\kappa}{2}\left(\frac{pD(p)}{hD(p)} - \mu\right)^2,$$

  optimal parameters are

$$\mu^* = \frac{\eta}{\eta - 1}, \qquad \kappa^* = \frac{1}{\eta}\left(\frac{\eta}{\eta - 1}\right)^{\eta} \frac{\sigma_\alpha^2}{\sigma_\varepsilon^2},$$

  where $\sigma_\alpha^2 = \mathbb{E}[\hat{\alpha}^2]$ and $\sigma_\varepsilon^2 = \mathbb{E}[\hat{\varepsilon}^2]$.

- κ^* depends on risk of manager discretion vs. value of private info, $\sigma_\alpha^2/\sigma_\varepsilon^2$.

- κ^* lower when η high, because market already limits influence of manager preference.

Pricing under optimal contract

- Manager sets

$$\hat{p} = \phi \left[\hat{c} + \frac{\eta^{\eta}}{(\eta - 1)^{\eta+1}} \hat{\alpha} \right] + (1 - \phi)\, \hat{h},$$

  where $(1 - \phi)$ is extent of accounting cost pricing:

$$1 - \phi = \frac{\sigma_\alpha^2 / \sigma_\varepsilon^2}{\sigma_\alpha^2 / \sigma_\varepsilon^2 + [(\eta - 1)^{\eta+1} / \eta^{\eta}]^2}.$$

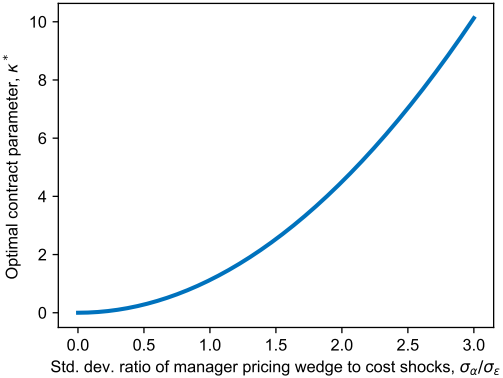
- Blended average of no-incentive price and accounting cost pricing.

- Extent of accounting cost pricing is increasing in $\sigma_\alpha^2 / \sigma_\varepsilon^2$ and decreasing in η.

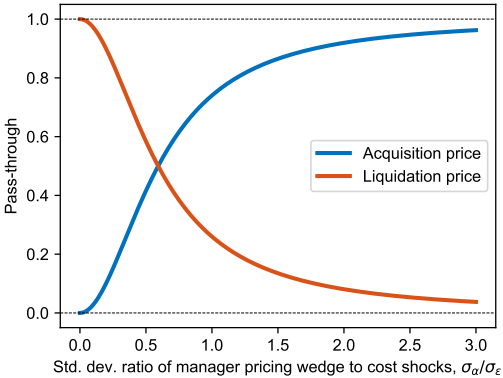
- Losses relative to first best, $\mathbb{E}[V^{\mathrm{FB}} - V]$, increasing in $\sigma_\alpha^2, \sigma_\varepsilon^2$, decreasing in η.

Pricing under optimal contract

- As risk of manager discretion grows, κ^* rises and ϕ falls.

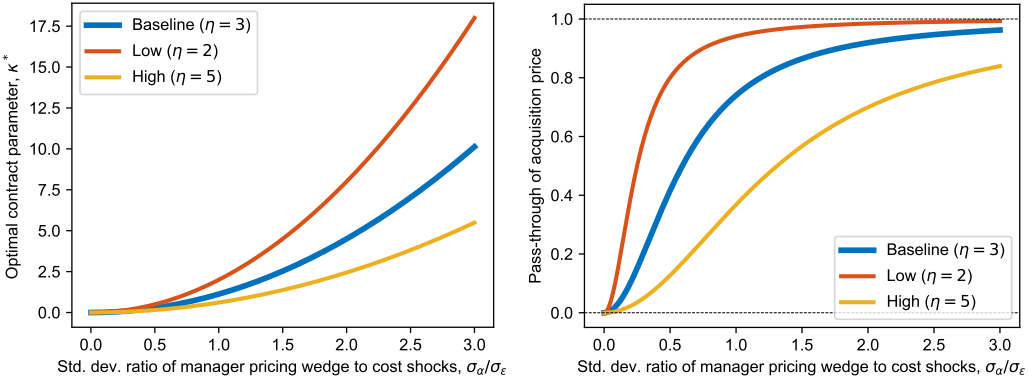


(a) Optimal contract parameter κ^*.



(b) Replacement vs. accounting cost pricing.

Accounting cost pricing strengthens when demand elasticity is low



(a) Optimal contract parameter κ^*.

(b) Weight on accounting cost pricing $(1 - \phi)$.

- The reliance of firms on historical costs is evident in many empirical studies. [....] The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal. (Okun 1981, pp. 156–157).

Partial cost observability

- Suppose there is a public signal of the replacement cost shock, $s \sim \mathcal{N}(\hat{\varepsilon}, 1/\tau)$.
- Principal can now condition payment on signal s:

$$K\left(pD(p), hD(p), s\right) = -\frac{\kappa}{2}\left(\frac{pD(p)}{hD(p)} - \mu(s)\right)^2.$$

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- Now, optimal contract parameters are:

$$\mu^*(s) = \frac{\eta}{\eta - 1}\exp\left(\frac{\tau\sigma_\varepsilon^2}{1+\tau\sigma_\varepsilon^2}s\right), \qquad \kappa^* = \frac{1}{\eta}\left(\frac{\eta}{\eta-1}\right)^\eta \frac{\sigma_\alpha^2}{\sigma_\varepsilon^2}\left(1+\tau\sigma_\varepsilon^2\right).$$

- Effectively, principal "marks-to-market" inventory value to $\exp\left(\frac{\tau\sigma_\varepsilon^2}{1+\tau\sigma_\varepsilon^2}s\right) hD(p)$.
- Lower posterior uncertainty makes incentives higher-powered.

Partial cost observability

- Manager sets

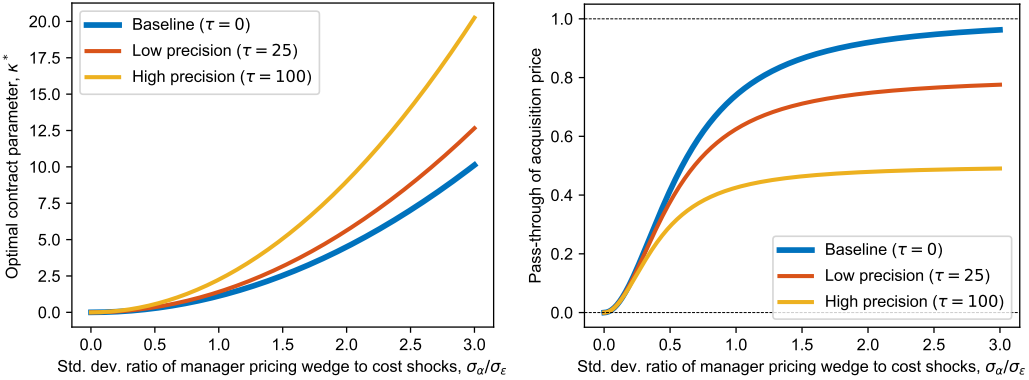
$$\hat{p} = \phi \left[\hat{c} + \frac{[(\eta-1)^{\eta+1}/\eta^{\eta}]^2}{[(\eta-1)^{\eta+1}/\eta^{\eta}]^2 + \tau\sigma_\alpha^2} \frac{\eta^{\eta}}{(\eta-1)^{\eta+1}} \hat{\alpha} \right] + (1-\phi)\,\hat{h},$$

  where $(1-\phi)$ is now

$$1-\phi = \frac{\sigma_\alpha^2/\sigma_\varepsilon^2}{\sigma_\alpha^2/\sigma_\varepsilon^2 + [(\eta-1)^{\eta+1}/\eta^{\eta}]^2 + \tau\sigma_\alpha^2}.$$

- Cost observability increases replacement cost pricing, but with stronger incentives.
- Also applies to perfectly observed shock (e.g., change in currency denomination).

Accounting cost pricing strengthens when cost observability is low



(a) Optimal contract parameter κ^*.

(b) Weight on accounting cost pricing $(1 - \phi)$.

- E.g., makers of gold/silver jewelry, typically use current market price of metal inputs.

Explaining other pricing puzzles

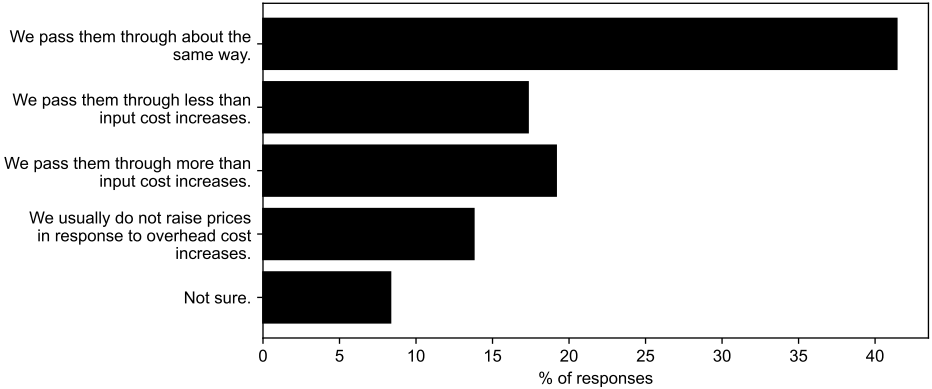
- Main result: Unverifiable info and principal–agent frictions mean:
  - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
  - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).

Explaining other pricing puzzles

- Main result: Unverifiable info and principal–agent frictions mean:
  - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
  - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).
- Fixed costs: Suppose both c and overhead cost F are stochastic.
  - Pass through innovations in F with same weight as $\hat{h}$, $(1-\phi)$.
  - Long history of survey evidence. Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
- Demand shocks: Suppose demand elasticity η is stochastic.
  - Pass through innovations in optimal markup with same weight as $\hat{c}$, ϕ.
  - Empirical evidence. Cagan (1979), Bils and Chang (2000), Fabiani et al. (2006), Gagnon and Lopez-Salido (2020), Kohler et al. (2026).

Firms' reported pass-through of fixed costs

- "Suppose your overhead costs go up, for example on rent, insurance, salaried staff or equipment leases. How do increases in such overhead costs affect your pricing—compared to increases in per-unit input costs on materials, wholesale goods, shipping per item, or hourly labor per job?"



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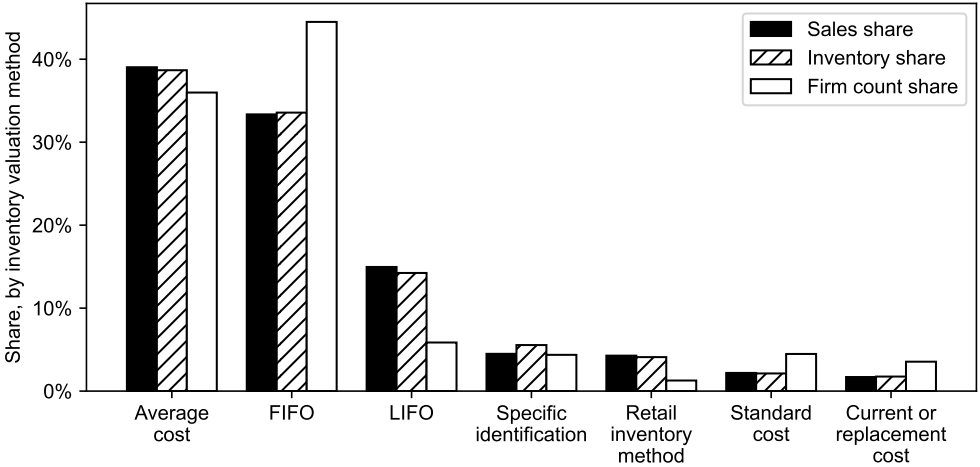
Calculating accounting costs

- In two-period setting, book value of inventory is unambiguous (*hl*).
- In dynamic setting, depends on how firm apportions cost across past purchases. E.g.,
  - Average acquisition cost (AAC).
  - First-in-first-out (FIFO).
  - Last-in-first-out (LIFO).
- Under all three, COGS reflects historical purchase costs. Methods differ in the weight they place on recent vs. more distant periods.
- $\Rightarrow$ Close connection between pricing with these inventory valuation methods and canonical sticky-price models.

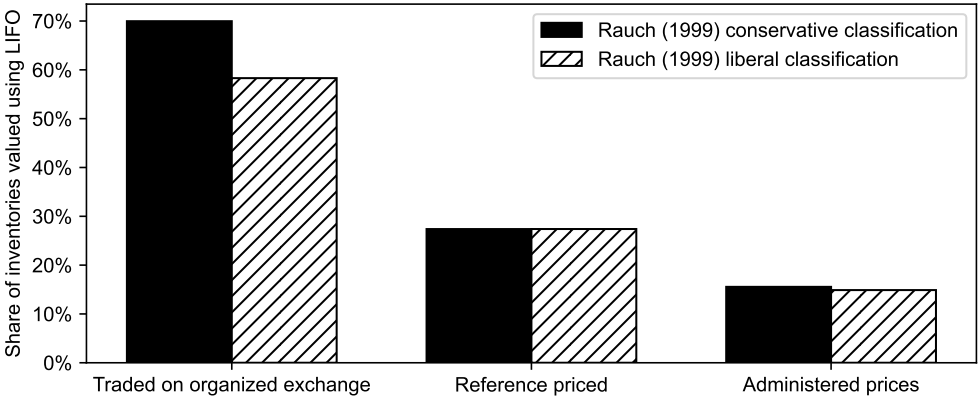
AAC and FIFO most common inventory valuation methods

- Valuation methods used by inventory-carrying public firms over 2010–2025:



AAC/FIFO more common among firms likely to price off accounting costs

- LIFO concentrated in markets with highly homogeneous outputs (organized exchanges) or inputs (reference-priced), where we expect less accounting cost pricing.



Price dynamics under average acquisition cost (AAC)

- Law of motion for inventories:

$$I_t = I_{t-1} - D_t + Y_t.$$

- Average acquisition cost (AAC) accounting:

$$h_t I_t = \left(I_{t-1} - D_t\right) h_{t-1} + Y_t c_t.$$

$\Rightarrow$ h_t is average of h_{t-1} and c_t, weighted by leftover inventory vs. new purchases.

- Log-linearizing around steady state with shipment-inventory ratio $\delta = D^{ss}/I^{ss}$,

$$\hat{p}_t = \hat{h}_t = (1-\delta)\,\hat{h}_{t-1} + \delta\,\hat{c}_t.$$

- Recursive formulation for prices looks familiar...!

Price dynamics under average acquisition cost (AAC) vs. Calvo

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{AAC}} = \delta \sum_{k=0}^{\infty} (1-\delta)^k \hat{c}_{t-k},$$

$$\hat{p}_t^{\text{Calvo}} = \theta \sum_{k=0}^{\infty} (1-\theta)^k \hat{c}_{t-k} + \underbrace{\theta \left[1-\beta(1-\theta)\right] \sum_{k=0}^{\infty} \sum_{s=0}^{\infty} \beta^s (1-\theta)^{k+s} \mathbb{E}_{t-k}\left[\hat{c}_{t+s-k} - \hat{c}_{t-k}\right]}_{\text{Due to forward-looking reset prices}},$$

  - AAC: δ is steady-state shipment-inventory ratio.
  - Calvo: θ is fraction of firms able to reset prices each period, and β is discount factor.

- Paths of prices coincide if $\delta = \theta$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s.

Price dynamics under average acquisition cost (AAC) vs. Calvo

$$\hat{p}_t^{\text{AAC}} = \delta \sum_{k=0}^{\infty} (1-\delta)^k \hat{c}_{t-k},$$

$$\hat{p}_t^{\text{Calvo}} = \theta \sum_{k=0}^{\infty} (1-\theta)^k \hat{c}_{t-k} + \underbrace{\theta \left[1 - \beta (1-\theta)\right] \sum_{k=0}^{\infty} \sum_{s=0}^{\infty} \beta^s (1-\theta)^{k+s} \mathbb{E}_{t-k} \left[\hat{c}_{t+s-k} - \hat{c}_{t-k}\right]}_{\text{Due to forward-looking reset prices}},$$

- AAC pricing behaves like “myopic Calvo” (Minton and Wheaton 2022).
- Not because of inattention, but because prices synchronized with accounting costs.
- Rate of decay determined by shipment-inventory ratio, not length of price spells.

Prices don't incorporate expected cost changes, but not b/c firms uninformed

- Fed Atlanta BIE survey: Firms' price changes incorporate realized, past cost changes but not expectations of future cost changes.
- Not because firms inattentive: Expectations predict future cost changes.

	Realized Price Change$_t$		*Realized Cost Change*$_{t+12}$	
	(1)	(2)	(3)	(4)
$\mathbb{E}_t$[Cost Change$_{t+12}$]	−0.082	−0.169	0.354**	0.294**
	(0.204)	(0.182)	(0.090)	(0.083)
Realized Cost Change$_t$	1.272**	0.979**	−0.050	−0.040
	(0.305)	(0.233)	(0.055)	(0.048)
Firm FEs	Yes	Yes	Yes	Yes
Date FEs	Yes		Yes	
Date-Sector FEs		Yes		Yes
N	3493	3480	1415	1410
R^2	0.48	0.56	0.65	0.73

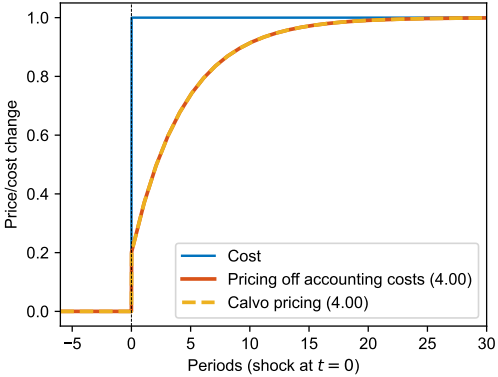
Differences in AAC vs. Calvo price dynamics

- Besides the fact that lags come from shipment-inventory ratios δ rather than price stickiness θ...

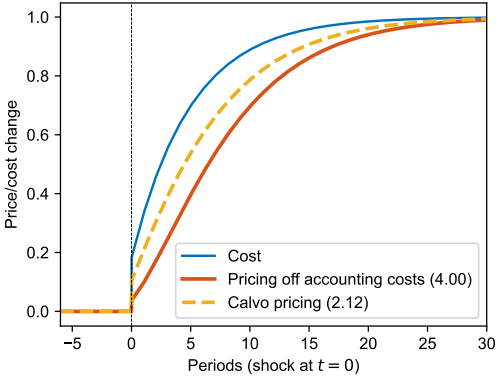
1. More delayed adjustment to slow-moving shocks.
2. Amplification of delays in supply chains.
3. Increased response to transitory shocks.
4. Delayed response when firms stock up before cost increase.

#1: More delayed adjustment to slow-moving shocks

- Set $\delta = \theta = 0.2$. Shock path, unexpected before $t = 0$: $\hat{c}_t = 1 - \rho^{t+1}$.



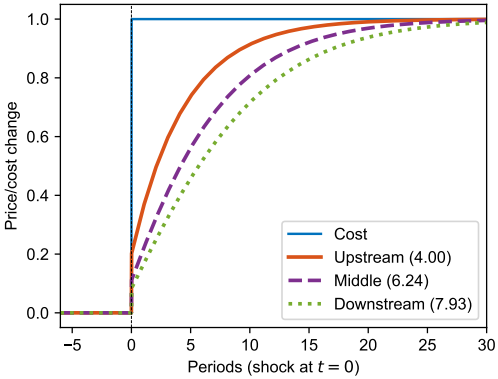
(a) Immediate shock ($\rho = 0$).



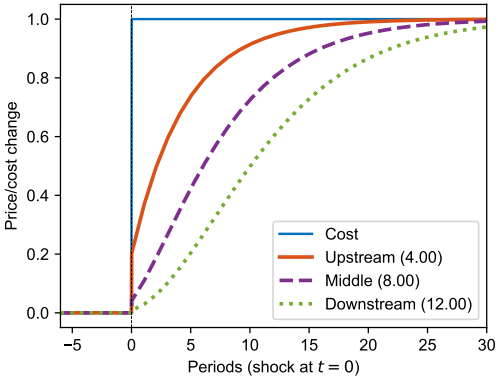
(b) Slow-moving shock ($\rho = e^{-0.2}$).

#2: Amplification of delays in supply chains

- Each firm turns cost shock into slower-moving shock for downstream firm.

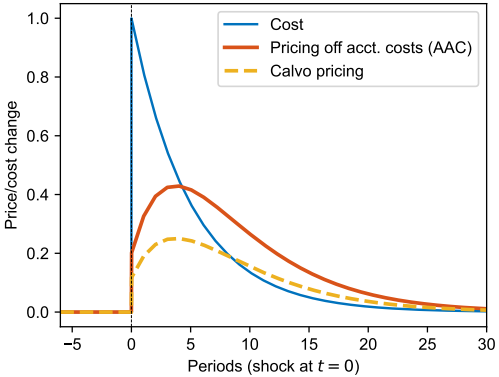


(a) Calvo.

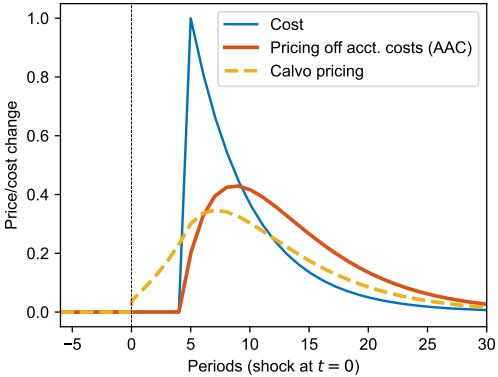


(b) Pricing off accounting costs (AAC).

#3: Increased responsiveness to transitory shocks



(a) Unanticipated cost increase at $t = 0$.



(b) News at $t = 0$ about cost increase at $t = 5$.

#4: Forward buying delays pass-through of cost increases

- First-order approximation for prices did not require keeping track of inventory stocks:

$$\hat{p} = (1-\delta)\hat{h}_{t-1} + \delta\hat{c}_t.$$

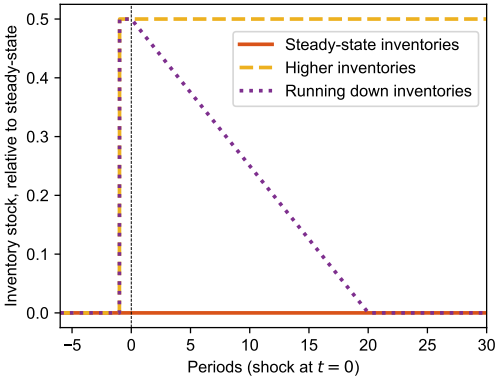
- Effects of stocking up on inventories show up at second-order:

$$\hat{h}_t = \underbrace{(1-\delta)\,\hat{h}_{t-1} + \delta\hat{c}_t}_{\text{First order}} + \underbrace{\frac{1}{2}\delta\,(1-\delta)\left(\hat{c}_t - \hat{h}_{t-1}\right)^2}_{\text{Second-order correction}} + \underbrace{\delta\left(\hat{c}_t - \hat{h}_{t-1}\right)\left(\hat{Y}_t - \hat{I}_t\right)}_{\text{Effect of inventories / purchases}}.$$

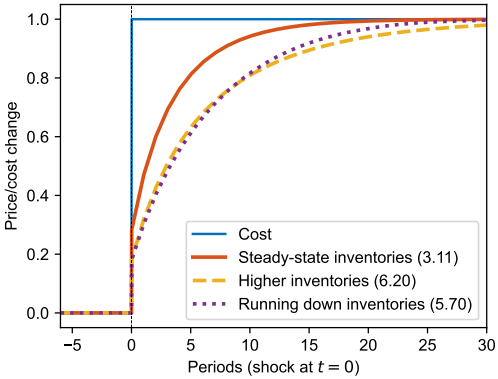
- If purchases low relative to inventory stock when replacement costs rise $\Rightarrow$ lower price.

#4: Forward buying delays pass-through of cost increases

- Illustrate with exogenous inventory paths in second-order approximation:

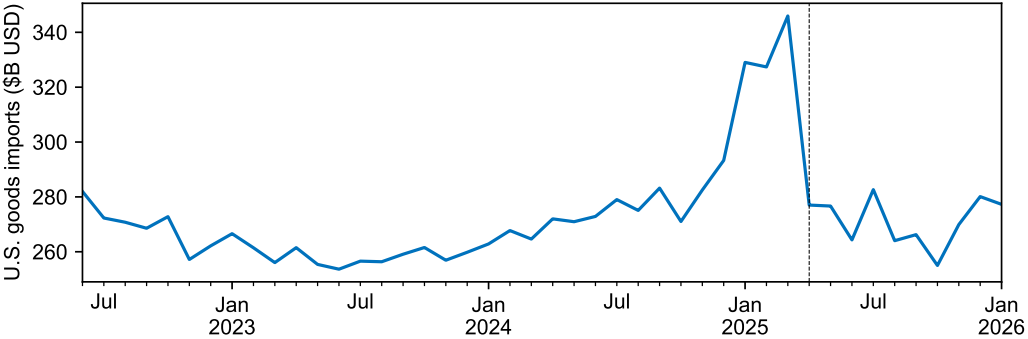


(a) Inventory paths.



(b) Pricing off accounting costs (AAC).

#4: Forward buying delays pass-through of cost increases



- Sticky prices: Need "myopic expectations" to amplify delays. Minton and Wheaton (2022).
- "The sensible firm just waits by the mailbox for news of cost increases and then, Okun-like, passes them on as price increases" (Gordon, 1990, p. 1150–1151).
- Accounting cost pricing + forward buying delays pass-through.

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$
$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$

$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k},$$

[Firms buy inventory every $1/\delta$ periods.]

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

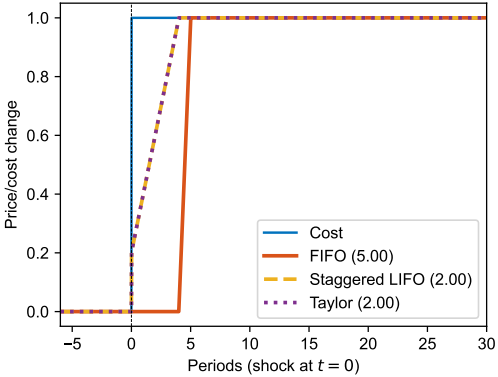
$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$
$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$
$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k}, \qquad \text{[Firms buy inventory every } 1/\delta \text{ periods.]}$$
$$\hat{p}_t^{\text{Taylor}} = \frac{1}{N}\sum_{k=0}^{N-1} \hat{c}_{t-k} + \underbrace{\frac{1}{N}\frac{1-\beta}{1-\beta^N}\sum_{k=0}^{N-1}\sum_{s=0}^{N-1} \beta^s \mathbb{E}_{t-k}\left[\hat{c}_{t+s-k} - \hat{c}_{t-k}\right]}_{\text{Due to forward-looking reset prices}}.$$

  - LIFO w/ intermittent purchases (LIFO-δ): Firms stock up on inventory every $1/\delta$ periods.
  - Taylor: Firm resets price every N periods, and β is discount factor.

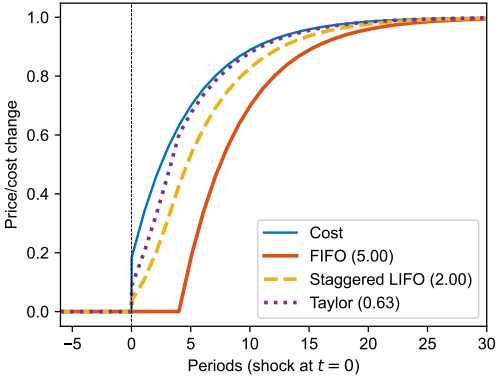
- LIFO-δ and Taylor coincide if $\delta = 1/N$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s.

FIFO, LIFO, and Taylor pricing

- Similar intuitions as AAC vs. Calvo.



(a) Immediate shock.



(b) Slow-moving shock.

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How Far Can We Go Without Sticky Prices?

Can we estimate the extent of accounting cost pricing?

- Theory predicts prices are weighted avg. of replacement- and accounting-cost prices:

$$\hat{p}_t = \phi \hat{c}_t + (1 - \phi)\hat{h}_t.$$

- Can we estimate ϕ?

Can we estimate the extent of accounting cost pricing?

- Theory predicts prices are weighted avg. of replacement- and accounting-cost prices:

$$\hat{p}_t = \phi \hat{c}_t + (1-\phi)\hat{h}_t.$$

- Can we estimate ϕ?

- Using $p_t \approx \mu(\phi c_t + (1-\phi)h_t)$:

$$\frac{\text{Sales}_t}{\text{COGS}_t} \approx \frac{\mu(\phi c_t + (1-\phi)h_t)D_t}{h_t D_t} = \mu\left(\phi \frac{c_t}{h_t} + (1-\phi)\right).$$

- $\phi > 0 \Rightarrow$ prices high relative to accounting cost when replacement cost high ($c_t > h_t$).

Can we estimate the extent of accounting cost pricing?

- How can we tell when $c_t > h_t$? Recall that:

$$\text{BookValueInventory}_t = \text{BookValueInventory}_{t-1} - \underbrace{\text{COGS}_t}_{h_t D_t} + \underbrace{\text{Purchases}_t}_{c_t Y_t}.$$

- We can construct:

$$\frac{\text{Purchases}_t}{\text{COGS}_t} = \frac{\Delta\text{BookValueInventory}_t + \text{COGS}_t}{\text{COGS}_t} = \frac{c_t Y_t}{h_t D_t}.$$

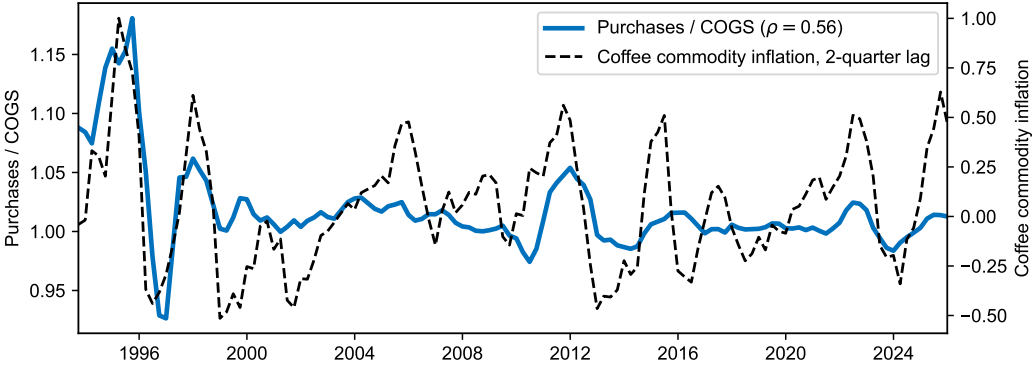
- So, log-linearizing around steady state:

$$\widehat{\left(\frac{\text{Sales}_t}{\text{COGS}_t}\right)} = \phi \widehat{\left(\frac{\text{Purchases}_t}{\text{COGS}_t}\right)} - \phi\left(\hat{Y} - \hat{D}\right).$$

- Absent changes in quantities and markups, ϕ identified by sensitivity of Sales/COGS to Purchases/COGS.

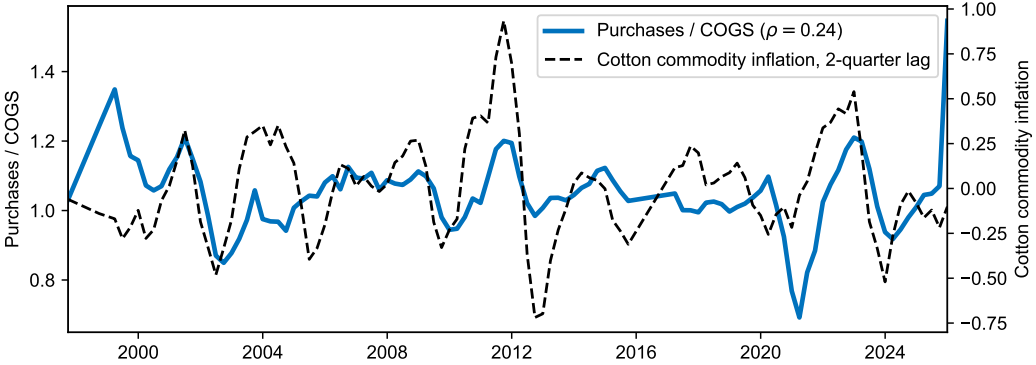
Does Purchases/COGS capture variation in c_t/h_t? Example: Starbucks

- Sales, COGS, Purchases $=$ ΔBookValueInventory $+$ COGS from quarterly reporting.
- Seasonality in Y_t/D_t: construct trailing four quarters cumulative variables.
- Compare to ICO Arabica coffee bean commodity price (via IMF).



Does Purchases/COGS capture variation in c_t/h_t? Example: Gildan

- Sales, COGS, Purchases $=$ ΔBookValueInventory $+$ COGS from quarterly reporting.
- Seasonality in Y_t/D_t: construct trailing four quarters cumulative variables.
- Compare to Cotlook benchmark cotton spot price (via IMF).



Estimating ϕ across all firms

- Quarterly financial reporting data for public firms in Compustat:
  - Exclude firms in information (NAICS 51), finance/insurance/real estate (NAICS 52–53).
  - Firms using AAC or FIFO. (Under LIFO, $h_t = c_t$.)
  - Observations with strictly positive sales, COGS, and inventory stock.
  - Change in total assets over past year $<$ 30% (avoid mergers / spin-offs).
- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

where α_{iq} are firm $\times$ quarter-of-year FEs.

Estimating ϕ across all firms

- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

- Change in relative quantities of purchases / shipments $\hat{Y}_t - \hat{D}_t$ is omitted variable:

$$\hat{\phi} \to \phi \left[1 - \frac{Var\left(\hat{Y}_t - \hat{D}_t\right)}{Var\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)} + \frac{Cov\left(\hat{h}_t - \hat{c}_t, \hat{Y}_t - \hat{D}_t\right)}{Var\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)} \right].$$

  - Seasonality: Firm $\times$ quarter-of-year FEs α_{iq}.
  - Long-term trends: "First differences" using $\log x_{it} - \log x_{it-4}$ for each variable x.
  - Attenuation bias: IV using NAICS-4 industry $\times$ Känzig (2021) oil shocks.

Estimating ϕ across all firms

- OLS estimates: $\phi \in [0.12, 0.15]$. Extent of accounting cost pricing $\approx$ 85%.
- Using NAICS-4 $\times$ Känzig (2021) oil shock as IV: $\phi \in [0.35, 0.4]$.
- Robust to control for Purchases/COGS of LIFO firms in industry (proxy for Y_t/D_t).

	Log(Sales / COGS)			
	Levels		First Differences	
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Log(Purchases / COGS)	0.147**	0.380**	0.126**	0.351**
	(0.015)	(0.104)	(0.014)	(0.079)
Firm $\times$ Quarter-of-Year FEs	Yes	Yes	Yes	Yes
N	382 211	369 469	345 468	333 935
R^2	0.86	0.86	0.09	0.04

Comparative statics: Replacement cost pricing in commodity industries

- Expect ϕ higher in industries with homog. outputs or observable costs. (Rauch 1999.)
- Expect ϕ higher in industries where more competitors use LIFO.

	Log(Sales / COGS)					
	Levels		First Differences			
	OLS	OLS	OLS	OLS	IV	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Log(Purchases / COGS)	0.122**	0.125**	0.101**	0.103**	−0.268	−0.136
	(0.016)	(0.019)	(0.017)	(0.016)	(0.376)	(0.265)
× Reference prices (r)	0.079**		0.074*		1.006*	
	(0.029)		(0.042)		(0.599)	
× Organized exchange (w)	0.254**		0.168**		0.720**	
	(0.062)		(0.051)		(0.329)	
× Industry LIFO share		0.185**		0.144**		0.744**
		(0.090)		(0.068)		(0.353)
Firm × Quarter-of-Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	251 379	276 804	227 964	251 033	225 546	242 882
R^2	0.88	0.88	0.09	0.09	0.08	0.01

Summary of estimates

- Average extent of accounting cost pricing $\approx$ 60–80%.
- Close to replacement cost in industries with reference prices (r) or exchanges (w).
- Accounting cost pricing $\geq$ 85% for firms in industries with administered prices.

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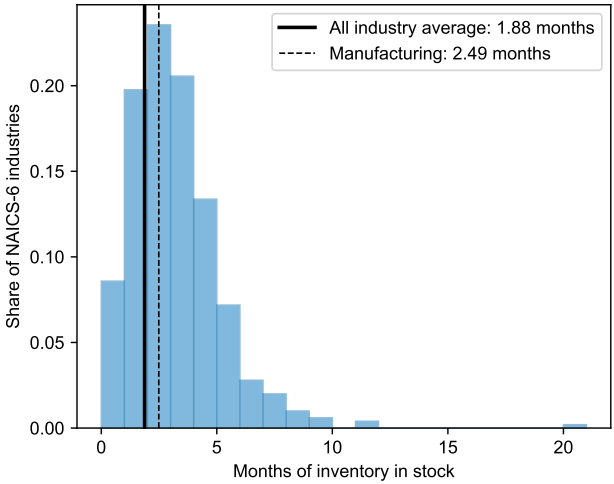
Quantitative input–output model of U.S. economy

Variable	Description	Source
Ω	Input–output matrix	BEA 2017 405-industry input-output table
β	Discount factor	$(0.96)^{1/12}$
θ_i	Frequency of price change	Pasten et al. (2020)
δ_i	Shipment-inventory ratios	Economic Census 2017 COGS / Inventories

- Benchmark input–output model with price rigidities á la Rubbo (2023).
- Inventories for mining (21), manufacturing (31–33), wholesale (42) & retail (44–45).
- “Hybrid” model where these industries have accounting cost pricing, rest have Calvo.
- Compare CPI response to commodity shocks to benchmark Calvo model.

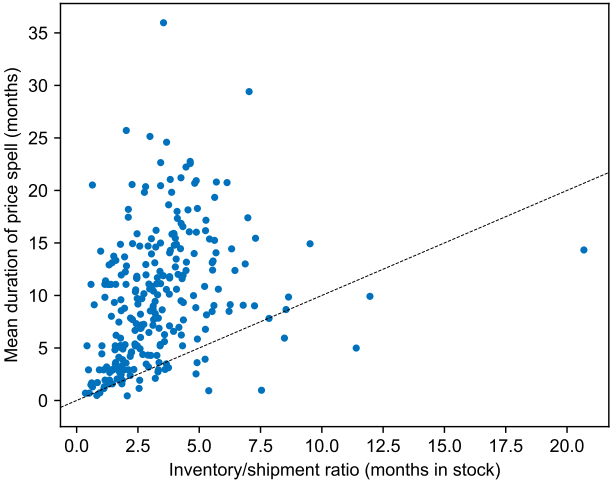
Inventory–shipment ratios from Economic Census



Industries with largest inventory–shipment ratios:

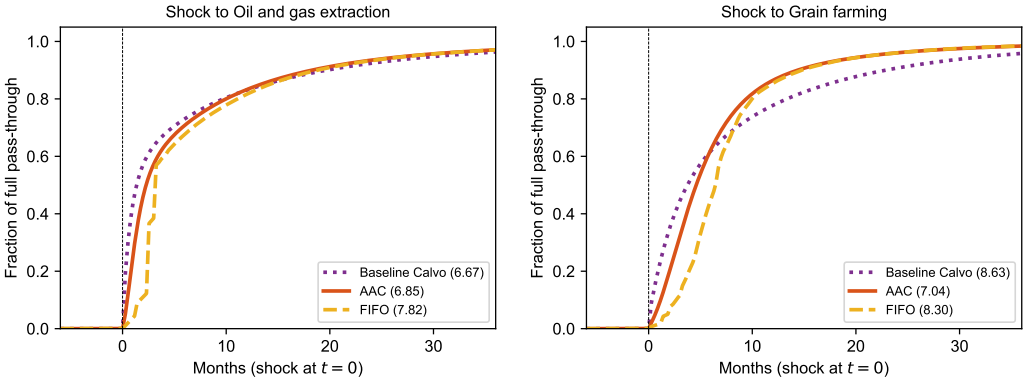
- Wineries.
- Biological product mfg.
- Aircraft mfg.
- Distilleries.
- Aerospace products and parts mfg.

Inventory–shipment ratios from Economic Census



- For most industries, $1/\delta_i < 1/\theta_i$ from Pasten et al. (2020).
- Two factors help accounting cost pricing.
- 1. Forward-looking reset prices under Calvo.
- 2. Reroute I–O table to account for margin industries.

Effects of commodity industry shocks on CPI



(a) Oil and gas extraction.

(b) Grain farming.

- Measure cumulative pass-through delay, $\sum_{t=0}^{\infty}(1 - \hat{p}_{\text{CPI},t}/\hat{p}_{\text{CPI},\infty})$.

Effects of commodity industry shocks on CPI

- Despite $1/\delta_i < 1/\theta_i$, accounting cost pricing generates similar delays in price adjustment to benchmark Calvo model.
- AAC or FIFO generate larger cumulative pass-through delay in about 1/2 of cases.

Industry description	Long-run pass-through (%)	Cumulative delay		
		Calvo	AAC	FIFO
Petroleum refineries	4.35	5.70	5.59	6.21
Electric power generation, transmission, and distribution	2.56	7.34	5.74	5.97
Animal (except poultry) slaughtering and processing	1.78	3.41	4.34	4.84
Other basic organic chemical manufacturing	1.53	13.25	10.99	11.82
Beef cattle ranching and farming	1.35	4.75	4.95	5.89
Grain farming	1.27	8.63	7.04	8.30
Plastics material and resin manufacturing	1.02	16.00	11.40	12.38
Iron and steel mills and ferroalloy manufacturing	0.87	18.37	12.10	13.15
Paperboard container manufacturing	0.80	13.55	9.89	10.68
Fruit and tree nut farming	0.79	6.05	5.98	6.91

Accounting cost pricing $\Rightarrow$ long and variable lags?

- "Industry experts estimate it takes at least nine months for raw bean prices to filter through to coffee drinkers, partly due to roasting times and price negotiations. [....] Most industry experts expect a coffee production surplus[...], which should, alongside the tariff removal, soften raw bean prices and eventually feed through to U.S. consumers. But this will take time, analysts say, because U.S. roasters typically hold about two to three months' worth of bean stocks on average and need another two to three months to roast and package their products."

  — Reuters, Dec 19 2025

- "The timing patterns observed among wholesale prices according to the stage of fabrication and from wholesale prices to consumer prices confirm the historical elements in costing. [...] It is reasonably predictable that today's changes in cattle prices will show up on the supermarket counter only after a lag; the butcher does not keep changing price tags in pace with changes in some putative replacement cost."

  — Okun (1981), *Prices and Quantities*

Conclusion

- Widespread reports that firms set prices using past inventory purchase costs.
- We develop a model of accounting cost pricing.
- Synthesizes other pricing puzzles:
  - Use irrelevant info: Hedging gains/losses, changes in fixed costs.
  - Neglect relevant info: Demand shocks, replacement costs, anticipated future costs.
- Devise approach to estimate extent of accounting cost pricing.
  - 60–80% overall, 85%–complete for firms with administered prices.
- Can resemble sticky-price models and generate similar delays in macro adjustment.

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Extra Slides

The Standard Theory: Firm Problem

- Given inventory I_{t-1} and current replacement cost c_t, firm value is:

$$V(I_{t-1}, c_t) = \max_{p_t, Y_t} \{p_t D_t - c_t Y_t + \beta \mathbb{E}[V_{t+1}(I_t, c_{t+1})]\},$$

where

$$I_t = I_{t-1} - D_t + Y_t. \qquad \text{(Inventory law of motion)}$$

$$mr_t(p_t) = D_t + p_t(dD_t/dp_t). \qquad \text{(Marginal revenue)}$$

- FOCs: Set marginal revenue = marginal cost = current replacement cost:

$$mr_t(p_t) = \beta \mathbb{E}\left[\frac{\partial V_{t+1}}{\partial I_t}\right] = c_t.$$

- Some (unsatisfactory) directions: $Y_t \geq 0$, $D_t \leq I_{t-1}$, perishable inventory, convex inventory carrying costs, borrowing constraint.

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